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# Optimization of laboratory-scale x-ray absorption spectroscopy (XAS) apparatus for nuclear fuel research

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## ABSTRACT

Dedicated to the characterization of mixed oxide nuclear fuels using x-ray absorption spectroscopy (XAS), the performance of a benchtop device was optimized. The setup is based on laboratory equipment, including a low-power x-ray tube and spherically bent crystal analyzers. By operating the device in the Johann asymmetric configuration, a significant performance enhancement is demonstrated: beam intensity increases by a factor of 3.16 at the uranium $L_3$-edge and 1.74 at the plutonium $L_3$-edge compared to previously reported symmetric configurations, while maintaining energy resolution between 0.6 and 2.8 eV in the 17–18 keV energy range depending on the crystal reflection used. This approach is readily extendable to other high-energy absorption edges of actinides, 4 d and rare earth metals. Using only 24 W of tube power and less than 24 h of acquisition time per sample, the device enables reliable uranium speciation. It holds promises for the speciation of plutonium as shown by x-ray absorption near edge structure spectra collected at the zirconium K-edge where Zr acted as a spectroscopic analog for Pu, with dilution limits expected to lie between 10% and 20%. Structural information can be obtained within one week of extended x-ray absorption fine structure measurements for concentrated samples. This device, named HotXAS, enables routine XAS characterization of radioactive materials in nuclear facilities, supporting future research on actinides, including the development of advanced nuclear fuels, the studies on environmental issues, and the long-term disposal of nuclear waste.

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## I. INTRODUCTION

X-ray absorption spectroscopy (XAS) is a well-established technique for characterizing the oxidation state and local structure of an element in a given compound, i.e., its speciation. Being a non-destructive and bulk technique applicable to solids, liquids, and gases, it is notably used throughout the nuclear fuel cycle, where the speciation of actinides and fission products plays an important role in the behavior of the fuel element. Indeed, XAS has been extensively used to characterize both conventional $UO_2$ and advanced $(U,Pu)O_2$ mixed-oxide (MOX) fuels for various purposes, such as monitoring manufacturing processes,[1,2] studying fuel behavior under normal[3–5] and accident conditions,[6,7] and assessing fuel behavior during storage.[8]

XAS experiments are primarily conducted at large user facilities, such as synchrotrons, where they benefit from extremely brilliant sources of x rays. However, one must deal with the oversubscription of these facilities when planning an experiment. The research process from the idea to the experiment can, thus, be very slow. This also excludes significant but routine experiments staying at the stage of beam time proposal. In the field of nuclear fuel and actinide research, the competition is tough as a limited number of beamlines can handle highly radioactive samples. Within Europe, these beamlines include ROBL at ESRF,[9] MARS at Soleil,[10] INE[11] and CAT-ACT[12] at ANKA. Taking into account the additional constraints involved in preparing radioactive samples and managing their transport to the beamline, it is obviously mandatory to develop complementary instruments to synchrotrons acting as


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valuable alternative to simplify access to x-ray absorption spectroscopy, in particular for routine measurements.

The specifications of the device when it comes to beam "production" are very straightforward considering the principle of the XAS experiment and desired data quality. The XAS experiment consists in illuminating a sample of thickness $t$ with an x-ray beam of energy $E$ and measuring the beam attenuation across the sample. The absorption cross section $\mu(E)$ is measured by collecting the intensity of the incident and transmitted beam, $I_0$ and $I_t$, respectively, and derived from the Beer–Lambert–Bouguer law, $I_t = I_0 e^{-\mu(E)t}$, for a given energy $E$. The measurement is then repeated as the energy of the beam is scanned across the absorption edge of the element of interest. Therefore, the device should first produce a beam of adjustable energy. Device performance should also be high enough to ensure desired quality of the collected spectrum. Spectral resolution is a first important metrics as it stands for the ability to resolve the spectral features ,such as pre-peaks or shoulder, in the x-ray absorption near edge structure (XANES) spectra. It depends greatly on the spread of energy in the incident beam (or instrumental resolution) as well as the core hole broadening, a physical limits independent of the device and inherent to the measured edge. The device designer aims to maximize the energy resolution to a point where it is at least better than the core hole broadening, which is about 8 eV at actinide $L_3$-edge energies to avoid further broadening of the spectral features. The signal-to-noise ratio (SNR) is another key metric. Seen in the smoothness of a spectrum, a high SNR enables weak features to be discriminated and reduces the error in the quantitative results obtained from spectrum analysis, such as oxidation state determination. Beyond increasing the acquisition time to improve the SNR, the designer must find a way to maximize the beam intensity while removing the sources of noise.

XAS experiments have always been carried out in laboratories. Since Fricke's pioneering work in 1920, when he measured the XAS spectra of metallic foils for the first time,[13] the development of laboratory instruments has been driven by the aim of making XAS easily accessible.[14–24] However, until recently, the performance of these instruments could not compete with that of synchrotrons. Improvements in the last decade to spherically[25,26] and cylindrically bent crystal analyzers (SBCA and CBCA, respectively), as well as in other types of curved crystal,[27] have greatly enhanced the flux and energy resolution achievable with those x-ray optics. This now allows synchrotron-quality data to be obtained in a few hours, or in half an hour for concentrated samples, making laboratory instruments complementary characterization tools.[28]

The vast majority of existing devices operate in the Johann configuration[29,30] or the von Hámos configuration,[31–33] utilizing spherical and cylindrical crystal analyzers as reflective optics, respectively. Although the von Hámos configuration enables data collection in a single-shot experiment, it typically has a lower collection solid angle (flux) per energy point than the Johann configuration. Devices employing either configuration generally operate efficiently within the 5–12 keV energy range and even between 2 and 5 keV installing a vacuum chamber to reduce air absorption, with notably applications for the measurement of actinides $M_4$ and $M_5$-edges as shown by the recent work of Yu *et al.*[34] demonstrating high count rates of about $10^5$ cps at U $M_4$ (3.7 keV) and $M_5$ (3.5 keV) and Th $M_5$ (3.3 keV)—edges with relatively low x-ray source power of 100 W. Flux reduction is, however, observed at higher energies, particularly in the 16–19 keV range of actinide $L_3$ edges, which are preferred to penetrate confinement barriers of radioactive samples.

The reduced performance of laboratory XAS setups at higher energies is partly due to the decreased scattering power of the diffracting plane of the crystal used to filter the photon energy, which is also referred to as reflection *hkl*, with *h*, *k*, and *l* standing for the corresponding Miller indices of the used diffracting plane. While the overall crystal efficiency, i.e., reflectivity, can be optimized by exploiting differences in the structure factor between odd and even reflections in commonly used Si and Ge crystals, this remains challenging to achieve in the Johann configuration. The choice of crystal reflection *hkl* in Johann devices has primarily been guided by accessibility of the energy range and energy resolution criteria. With the need to minimize Johann error, reflectivity was a secondary consideration. However, after only one use in a flat dispersive laboratory XAS setup,[35] asymmetric configuration is now supported by Johann type devices,[36] allowing for greater flexibility in operation. A single crystal can now support multiple reflections while simultaneously reducing the Johann error.[37,38]

Initial research in the hard energy range explored various device configurations. The Mo K-edge (20 keV) was measured using a Laue–Dumond type spectrometer[39] and a double-crystal configuration,[20] while the Rh K-edge (23.2 keV) was also measured with the same double-crystal configuration.[20] The Zr K-edge (18 keV) was collected using a transmission dispersive arrangement[40] and more recently with a von Hámos spectrometer.[41] In the case of actinides, the feasibility of conducting XAS experiments, including nuclear fuel samples, has also been demonstrated. The Johann configuration with reflexive optics (SBCA) yielded promising results at the U $L_3$-edge (17.2 keV). Bes *et al.* measured $UO_2$, $UO_3$, and $KUO_3$ samples[42] within 24 h. Shortly thereafter, Jahrman *et al.* measured $(PPh4)_2UCl_6$[43] over 46 h, followed by Mottram *et al.*, who measured $UTi_2O_6$, $U_{0.55}Yb_{0.25}Ti_2O_6$, $UO_3$, and $UTi_2O_6GC$[44] samples in 15 h. More recently, Novichkov *et al.* measured the $L_3$-edge of actinides, such as Th (16.3 keV) in $ThO_2$ within 13 h, U in $UO_2$ in 9 h, Np (17.6 keV) in $NpO_2^+$, and Pu (18.06 keV) in $PuO_2^{2+}$ within 66 h.[45] Looking at the acquisition time of these measurements, improvements in flux is still highly desirable, especially at the Pu $L_3$-edge.

This paper addresses this issue by introducing the HotXAS device, which is primarily designed to handle radioactive samples in a nuclear research facility. First, practical details regarding the choice of device components for XAS experiments at actinide $L_3$ energies are provided. The paper then discusses energy resolution and flux optimization in the energy range of U $L_3$ and Pu $L_3$-edges based on operation in the asymmetric Johann configuration. Finally, the sensitivity of the device for measuring the U oxidation state is evaluated as well as its potential for measuring extended x-ray absorption fine structure (EXAFS), demonstrating its suitability for the routine characterization of nuclear fuel samples.

## II. DESIGN OF THE DEVICE

### A. Building the spectrometer

The HotXAS device is designed to handle radioactive solid samples for characterizing as manufactured MOX fuels samples,

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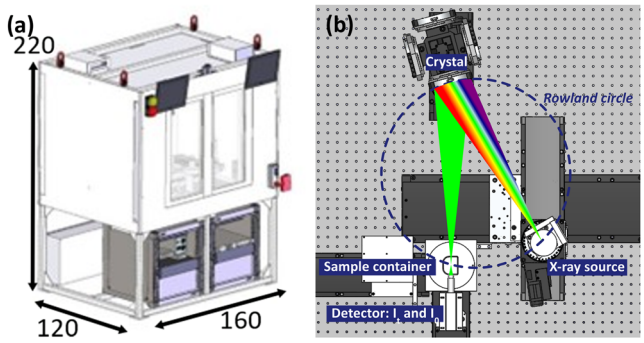



**FIG. 1.** HotXAS device equipment: shielding unit (a) and components layout (b).

containing actinides, primarily uranium and plutonium, as well as simulated irradiated MOX fuels (SIM-MOX), which incorporate stable fission products.[5,46] Due to the high toxicity of these samples, which are a source of contamination and irradiation (particularly for the samples with the highest Pu content) under manipulation, confinement and screening measures were implemented. Currently, a sample container is being developed to handle as-manufactured sample pellets, ensuring both confinement and screening (limiting the dose rate for samples with the highest Pu content). The tests reported in this paper were performed on uranium compounds that posed no risk of irradiation, for which normal synchrotron confinement was used. Zirconium compounds were also measured, with Zr acting as a spectroscopic surrogate for Pu and not requiring specific handling.

The device is compact enough to be installed inside a nuclear laboratory. It is enclosed in a shielding unit with a footprint of 220 × 160 × 120 $cm^3$ as depicted in Fig. 1(a). Inside the radioprotection unit, the instrumentation was designed around the sample environment, as depicted in Fig. 1(b). It consists of three components placed on a circle (the so-called Rowland circle): an x-ray source, a crystal analyzer, and a detector. First, the x-ray source generates a white beam that illuminates the crystal. The crystal then selects the desired energy, with its bent shape focusing the monochromatic beam on the sample. At the end of the beam path, the detector measures the transmitted signal $I_t$. The intensity of the incident beam $I_0$ can be easily measured by simply removing the sample from the beam. Slits are placed in front of the sample to prevent scattered x rays from reaching the detector directly.

The shielding unit manufactured by Lemer Pax Inc. (France) provides total protection from the x-ray source up to energies of 60 keV (corresponding to the maximum accelerating voltage of the x-ray source power supply). The unit has two sliding doors at the front for accessing the device during operation (e.g., changing the sample) and two doors at the back for the maintenance (e.g., loading a new crystal). An interlock shutting down the tube power is connected to all the doors. The unit also has four x-ray labyrinths for the cables of the motors, tube, and detectors.

## B. Operating the device

The HotXAS device operates in the Johann configuration. In this geometrical configuration, the device components move along the Rowland circle during spectrum acquisition to scan the incident energy as represented in Fig. 2(a). The Bragg law defines the relationship between the energy $E$ and the Bragg angle $\theta$, based on the d-spacing $d_{hkl}$ of the $hkl$ crystal plane involved in the reflection,

$$2d_{hkl}\sin(\theta) = \frac{hc}{E}, \tag{1}$$

where $h$ and $c$ are the Plank constant [kg $m^2$ $s^{-1}$] and the speed of light [m $s^{-1}$], respectively. In the HotXAS device, the sample remains in a fixed position to comply with nuclear safety regulations. As the device is also designed to operate in fluorescence mode, adaptation of the Johann configuration with a moving x-ray source was implemented. This fixed sample configuration also facilitates the installation of instrumentation for *in situ* XAS experiments. During a scan from low to high energy, the device transitions from high to low Bragg angles as shown in Fig. 2. Initially, the tube is near the sample and the crystal is at its farthest point, positioned almost

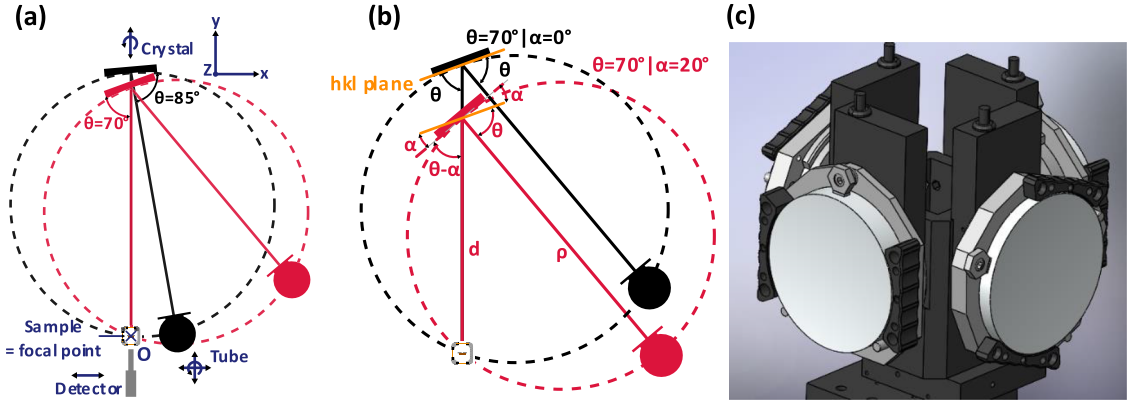



**FIG. 2.** Movements of the components switching the Johann configuration from 85° to 70° Bragg angle in symmetric operation (a) and movements of the components switching the Johann configuration from symmetric to asymmetric operation at 70° Bragg angle and 20° asymmetry angle (b). CAD of the crystal revolver with its azimuthal rotations (c).

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perpendicular to the Oy axis. As energy increases and Bragg angle decreases, the tube moves away from the sample, rotating along the Oz axis and shifting along both Ox and Oy axes, while the crystal approaches the sample, rotating and moving along the Oz and Oy axes. The sample remains stationary at the meridional focal point. Another translation stage pulls away the detectors along the Ox axis for loading and unloading the sample container. The device is equipped with translations and rotations from Newport Inc. (USA).

The HotXAS device can operate in both the symmetric and the asymmetric Johann configurations. Figure 2(b) shows how the geometry changes when switching from the symmetric to the asymmetric configuration. In the symmetric configuration, the diffracting plane *hkl* is parallel to the crystal surface. The incident and exit angles are then equal to the Bragg angle. By rotating the crystal through an asymmetry angle $\alpha$, a different hkl plane enters diffracting conditions in the asymmetric configuration. In this configuration, the incident angle increases while the exit angle decreases by the asymmetry angle, resulting in a corresponding increase and decrease in chord lengths $\rho$ and $d$,

$$\rho = D \sin(\theta + \alpha), \tag{2}$$

$$d = D \sin(\theta - \alpha), \tag{3}$$

where $D$ denotes the diameter of the Rowland circle.

The accessible energy range, or Bragg angle range, for a given reflection depends on the range of motion of the motorized stages setting the lower Bragg angle limit, or highest energy, as well as the footprint of the sample environment setting the lower Bragg angle limit, or lowest energy. In symmetric configuration, the device scans a Bragg angle range from 61.5° up to 86.5° thanks to the optimized footprint of the sample container, enabling a 2 keV energy range coverage at actinide $L_3$ edge energies. For example, the Ge[9 9 9] reflection from the Ge[1 1 1] crystal scans an energy range from 17.11 to 19.45 keV covering U (17.17 keV), Np (17.61 keV), Pu (18.06 keV), Am (18.51 keV), and Cm (18.97 keV) $L_3$ XANES with Bragg angles of 84.3°, 75.9°, 71.1°, 67.3°, and 64.2° at the mentioned edges and even Th (16.30 keV) and Pa (16.73 keV) $L_3$ using the Ge[8 8 8] reflection yielding 68.7° and 65.1° Bragg angles at the edge energies, respectively. However, due to geometrical reflections errors, mostly the Johann error increasing the acceptance angle and being potentially detrimental for XANES analysis, one crystal is usually dedicated to one restricted energy range where the Bragg angle is closer to 90° to increase the energy resolution. While EXAFS analysis accepts some decrease in energy resolution, it requires extended scans in k space and, thus, also benefits from higher Bragg angle reflections at the element edge energy. Considering both XANES and EXAFS requirements, element versatility in symmetric configuration comes from the diversity of the crystal orientations. Taking the same example, Th, Pa, U, Np, Pu, and Am would be better measured with Si[10 10 0], Si[12 8 0], Si[12 8 4], Si[9 9 9], and Si[16 0 0], yielding Bragg angles of 82.1°, 79.6°, 84.4°, as well as 76°, 80.2°, and 80.7° at the mentioned edges. To fasten crystal changes and, thus, edge changes, the device is equipped with a crystal exchanger, which can be loaded with up to four pre-aligned crystals as depicted in Fig. 2, useful for multi-edge experiments.[47]

In asymmetric configuration, the accessible lower Bragg angle reduces to 68.5° and 78.5° at 10° and 20° asymmetry, respectively, due to the shortening of the chord length $d$, while the higher Bragg angle is unchanged. Despite the Bragg angle range in the present case being further limited using the same motor positioning, this configuration allows greater versatility. The energy range scanned by a given crystal can be extended by several keV performing "*hkl* hopping." When reaching either mechanical or energy resolution limits with one reflection, another reflection plane with a lower d-spacing and, thus, higher Bragg angle brought into diffraction conditions by adjusting the asymmetry angle, allowing access to a higher energy range.[37] This configuration also allows the incident angle to be brought closer to 90°, choosing the reflection and the asymmetry angle appropriately,[36] enhancing the energy resolution compared to the symmetric configuration in a procedure called Johann Normal alignment.[38] In the given example, all $L_3$ edges XANES from Th to Am can be measured with the Si[5 5 1] crystal yielding Si[10 10 0], Si[11 9 3], Si[10 10 4], Si[11 9 5], Si[11 11 1], Si[13 9 3], and Si[11 11 5] reflections, respectively, at Bragg and asymmetry angles of (82.1°, 8°), (82.4°,6.8°), (77.8°, 7.7°), (77.6°, 12.6°), (80.3°, 4.4°), (83°, 10.5°), and (79.6°, 9.8°). In practice, reflection changes are performed with the crystal motor rotation that sets the incidence angle $\theta_M = \theta + \alpha$. This also requires the adjustment of the azimuthal angle of the crystal, which is achieved using an additional motorized rotation stage mounted on each crystal slot, as shown in Fig. 2.

## C. Choice of the device components and optimization at actinide $L_3$-edge energies

### *1. Choice of the crystal reflections for simultaneous flux and resolution optimization*

The selection of crystal reflection for scanning U and Pu $L_3$-edges follows a novel approach in this work, showcasing an additional advantage of the asymmetric configuration. This approach not only considers the conventional criteria of energy resolution and edge accessibility, as is typical in the symmetric configuration, but also incorporates reflectivity as an essential parameter. By optimizing for higher detected incident intensity ($I_0$), this approach aims to enhance measurement statistics or reduce the acquisition time.

This method relies on the increase in the number of accessible reflections to choose from using the asymmetric configuration compared to the symmetric configuration. For instance, when measuring the U and Pu $L_3$-edges with a set of eight crystals available in the spectroscopy center (listed in Table I and manufactured at the European synchrotron radiation facility (ESRF) and by XrsTech Inc. (USA) with a 0.5 m bending radius and 150 $\mu$m thickness), the symmetric configuration yields only eight possible reflections for each edge, with mechanical limits often reducing this to two, and ultimately, only one is selected due to resolution requirements. By contrast, the asymmetric configuration enables the same set of eight crystals to provide between 60 and 80 possible reflections (combination) under Johann Normal Alignment (JNA) conditions and within the accessible mechanical range. On average, this results in ~10 reflections per crystal at actinide energies.

The detected incident intensity ($I_0$) depends not only on the reflectivity but also on other factors. Some of these factors are independent of the chosen reflection and relate to the instrumentation settings. These include the x-ray tube power ($P_{tube}$), accelerating voltage, filament intensity, the detector quantum efficiency ($\varepsilon_d$),

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**TABLE I.** List of available crystal integrated reflectivity $F_{hk}$ for suitable asymmetric reflections to maximize the detection efficiency DE at U and Pu $L_3$-edges alongside commonly used symmetric reflections highlighted in boldface. The expected gain is reported as a ratio between the detection efficiency given by the asymmetric reflection and the commonly used symmetric reflection, and compared to measured ratio.

| $G_{hkl}$ | Crystal | Θ (°) | $\alpha$ (°) | $\Theta_M$ (°) | $F_{hkl}$ (meV) | $CE$ | $DE$ (meV) | $r_{sim}$ | $r_{meas}$ | $I_0$ (cps) |
|---|---|---|---|---|---|---|---|---|---|---|
| **U $L_3$: 17 166 eV** | | | | | | | | | | |
| Si[12 8 0] | Si[5 3 1] | 73.6 | 10.1 | 83.7 | 4.97 | 0.78 | 3.88 | 2.23 (0.29) | 2.24 (0.02) | 502 |
| Si[12 6 6] | Si[5 3 3] | 77.6 | 5.1 | 82.8 | 5.34 | 1 | 5.34 | 3.09 (0.36) | 2.43 (0.01) | 544 |
| Si[10 10 4] | Si[3 3 1] | 77.8 | 2.5 | 80.3 | 5.10 | 1 | 5.10 | 2.99 (0.38) | 3.16 (0.02) | 708 |
| Si[10 10 4] | Si[5 5 1] | 77.8 | 7.7 | 85.5 | 5.49 | 1 | 5.49 | 3.15 (0.36) | 2.67 (0.02) | 599 |
| Si[12 8 4] | Si[6 4 2] | 84.5 | 0 | 84.5 | 4.17 | 1 | 4.17 | 2.40 (0.33) | 2.73 (0.01) | 613 |
| *Ge[9 9 9]* | ***Ge[1 1 1]*** | ***84.3*** | ***0*** | ***84.3*** | **1.74** | **1** | **1.74** | ··· | ··· | ***224*** |
| **Pu $L_3$: 18 057 eV** | | | | | | | | | | |
| Ge[14 6 4] | Ge[4 2 2] | 72.9 | 9.9 | 82.8 | 3.53 | 0.74 | 2.61 | 1.34 (0.17) | 1.40 (0.02) | 256 |
| Ge[10 10 8] | Ge[1 1 1] | 80.3 | 5.8 | 86.1 | 2.86 | 1 | 2.86 | 1.45 (0.19) | 1.74 (0.02) | 328 |
| Si[14 6 0] | Si[5 3 1] | 74.4 | 12.4 | 86.8 | 3.35 | 0.77 | 2.58 | 1.31 (0.24) | 1.43 (0.02) | 269 |
| Si[12 10 2] | Si[5 5 1] | 84.7 | 5.2 | 89.8 | 3.33 | 1 | 3.33 | 1.68 (0.29) | 1.74 (0.02) | 328 |
| *Si[9 9 9]* | ***Si[1 1 1]*** | ***80.2*** | ***0*** | ***80.2*** | **1.92** | **1** | **1.92** | ··· | ··· | ***183*** |

and the filter thickness ($t_{Al}$). On the other hand, the other factors are related to the chosen reflection in the form of geometric factors: the crystal's solid angle relative to the x-ray source ($SA_c$) and the detector's collection efficiency ($CE_d$), with the Bragg angle and asymmetry angle affecting both the chord length ($\rho$), as seen in equation,[2] and the beam's spread on the finite-size detector, considering the SBCA's astigmatism, which increases as the Bragg angle decreases and the asymmetry angle increases.[38] This can be summarized as follows:

$$I_0 \propto P_{tube} SA_c F_{hkl} e^{-(\mu_{Al}\rho_{Al}t_{Al})} CE_d \varepsilon_d, \tag{4}$$

with $\mu_{Al}$ and $\rho_{Al}$ being the total photon attenuation coefficient at the scanned energy ($cm^2\ g^{-1}$) and the density of aluminum ($g\ cm^{-3}$), respectively.

In a benchmark study aimed at maximizing the intensity of the incident beam, we set instrumentation parameters at a fixed value (tube power, filter thickness, and detector settings) and selected several reflections to maximize the double product $F_{hkl}CE_d$ while maintaining the instrumental energy resolution to the required level. Four asymmetric reflections were chosen for the U and Pu $L_3$ edges, respectively, from six different crystals. Table I gives the corresponding Bragg angle $\theta_B$, asymmetry angle $\alpha$, mean incident angle $\theta_M$ ($\theta_B + \alpha$), and integral reflectivity $F_{hkl}$ computed with the PyTTE software.[48] For comparison, the best symmetric reflections were selected from commercially available crystals based on the resolution criterion solely: Si[12 8 4] from Si[6 4 2] and Ge[9 9 9] from Ge[1 1 1] for the U $L_3$-edge and Si[9 9 9] from Si[1 1 1] for the Pu $L_3$-edge. The size of the beam at the detector position was evaluated by x-ray tracing simulations to compute the collection efficiency (CE). The crystal solid angle was estimated at 32 msr and is relatively constant for all the reflections given the constant mean incident angle $\Theta_M$. The double product $F_{hkl}CE_d$ is also computed in Table I and is referred to as detection efficiency (DE), an overall indicator of the detected intensity given by a reflection when all the other device parameters are set.

### 2. X-ray source

The choice of x-ray source was driven by mechanical integration constraints as well as accessibility of the energy range. X-ray tubes provide a compact source of x rays that can be integrated into laboratory XAS devices. They produce a continuous x-ray spectrum with an energy range covering both soft and hard x-ray energies, as well as the characteristic lines of the anode and cathode materials. From a mechanical point of view, the HotXAS device configuration with a fixed sample also requires the tube to move, so weight is also a consideration for motorization.

During the early development phase, the HotXAS device was equipped with a micro-focused x-ray tube from Micro X-Ray Inc. (USA). The tube is cooled using fans mounted on its support and operates at a maximum power of 50 W. As can be seen in Fig. 1, its compactness and light weight make integration into the moving-tube system easy. Newport stages can support heavier, higher power tubes, such as water-cooled x-ray tubes, thanks to a 22 kg load capacity across the accessible Bragg angle range, while maintaining fast positioning. However, additional engineering work is needed to confirm potential upgrades, including ensuring cooling hose integrity during operation. The micro-focused tube was equipped with a palladium anode to avoid x-ray characteristic lines in the 16–19 keV energy range. Above 19 keV, the palladium Kα emission lines appear strongly in the detector signal when accelerating voltage exceeds 24 kV due to inelastic scattering. Although this is not relevant for the measured energy range in the present work, accelerating voltage and filament current were kept at 24 kV and 1 mA to prolong x-ray tube lifespan. The tube has a source size of 75 $\mu$m, ensuring a low contribution of the source size to the instrumental resolution.

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### 3. Detector

*a. Choice of the detector* While an ionization chamber can be used on a synchrotron beamline to measure beam intensity, laboratory XAS devices require an energy selective detector due to the presence of reflection harmonics. With the exception of one reported device using a 12 kW tube,[20] most laboratory configurations employ a single optical element to generate the incident beam. This beam contains undesirable photons resulting from inelastic scattering and harmonics, which must be eliminated or at least attenuated. In the actinide $L_3$ energy range (16–19 keV), these undesirable photons present a challenge to the detector. First, the lower order harmonics are generally much more intense than the high order harmonics and must be filtered to prevent the detector from saturating. As they cannot be eliminated, the detector resolution should be better than 1 keV in order to separate them. Second, under certain conditions, the inelastic background can become a significant contribution to the detected signal, leading to increased noise and, in the worst case, distortion due to thickness effects.[49] The most unfavorable circumstances arise when a high inelastic cross section is combined with low crystal integral reflectivity, as encountered in high-energy experiments, particularly when coupled with limited detection energy resolution and insufficient shielding. For such cases, the background must the measured separately and subtracted from the signal at each energy step in order to compute the absorption cross section $\mu$[30],

$$\mu \propto \ln\left(\frac{I_0 - I_{0,bgk}}{I_t - I_{t,bgk}}\right), \tag{5}$$

where $I_{0,bgk}$ and $I_{t,bgk}$ stand for the intensity of the background in the incident and transmitted beam signals, respectively. For MOX samples, the presence of Pu and Am isotopes results in supplementary background from x-ray and gamma emissions associated with radioactive decay. This increased background lead to the selection of a detector with high count rate capabilities.

The HotXAS device incorporates two detectors. In the 16–19 keV energy range, the X-123-CdTe detector from Amptek Inc. (USA) is preferably used, as its 1 mm-thick CdTe sensor guarantees 100% quantum efficiency. With a peaking time of 2.4 µs, it supports a count rate of $1.2 \times 10^5$ cps, achieving an energy resolution of 530 eV at 14.4 keV. Its active detection area, which is 5.6 mm in diameter, ensures 100% collection efficiency from 90° to 77.5° in symmetric configuration, although performance is reduced in asymmetric configuration, as discussed in detail by Chen *et al.*[35] For the HotXAS device configuration in particular, the impact of the Bragg angle, asymmetry angle, sample size, and detector size on collection efficiency was evaluated in detail in Figs. S1 and S2 of the supplementary material. When a larger detection area is required, the VITUS H150 SDD detector from Ketek Inc. (Germany) is preferred. Its 13.8 mm diameter active area ensures 100% collection efficiency from 90° to 73.5°. Its 450 $\mu$m-thick silicon sensor provides a quantum efficiency of 52% at 18 keV. With a peaking time of 0.1 $\mu$s, it has an energy resolution of around 136 eV at 6 keV and a maximum count rate of $2 \times 10^6$ cps.

*b. Background attenuation* The detection system was optimized to further reject inelastic scattering and attenuate radioactive background and lower-order harmonics. Taking advantage of their angle distribution, thin rectangular tungsten slits were placed at the focal point to block the incoherently scattered rays. The impact of the slit opening on the inelastic background contribution and SNR is reported in Fig. S3 of the supplementary material. At the U $L_3$-edge, the background accounts for 11% of the total incident intensity collected with the Si[12 8 4] reflection using $2 \times 16$ mm$^2$ opening, compared to 25% with $8 \times 16$ mm$^2$ opening. This enhances the SNR by 6%, using the same acquisition parameters and device settings. At the Pu $L_3$-edge, the background contribution is about 5% using the Ge[10 10 8] reflection and $2 \times 16$ mm$^2$ slits. Given these low contributions, no background subtraction was necessary to process the $I_t$ and $I_0$ data collected at the Zr K and U $L_3$ edges. The $2 \times 16$ mm$^2$ slits and the positioning of the detector 25 mm away from the sample also reduce the radioactive background. A simplified model was used in the supplementary material, with model parameters given in Table S1 and results for various Pu contents in Tables S2–S4. Primary x-ray and gamma emissions following nuclear decay[50] generate at most $10^5$ cps and $1.5 \times 10^4$ cps, resulting in an additional 2.2 and 6% dead time for the Si-SDD and CdTe detectors, respectively, when analyzing the samples with the highest Pu content.

Lower order harmonics were addressed using aluminum filters. As the filter also reduces the intensity of the harmonic of interest, a thickness of 300 $\mu$m was found to be an effective compromise, attenuating 89% of photons below 10 keV while allowing 64% of photons above 17 keV to pass through. When germanium crystals are used, the aluminum filter also attenuates the very strong Ge Kα (9.9 keV) and Kβ (11 keV) fluorescence lines. Regarding background attenuation, Fig. 3 shows another advantage of the asymmetric configuration for generating harmonic-free beam by accessing easily to diffracting planes with high [*h k l*] fundamental. At the U $L_3$-edge, the Si[10 10 4] or Si[12 8 0] reflections yield a monochromatic beam, free from higher order harmonics for accelerating voltage below 34 kV, as opposed to the Si[12 8 4], Si[12 6 6] or Ge[9 9 9] reflections, which come inevitably with lower order harmonics. At the Pu $L_3$-edge, all the selected asymmetric reflections are also free from

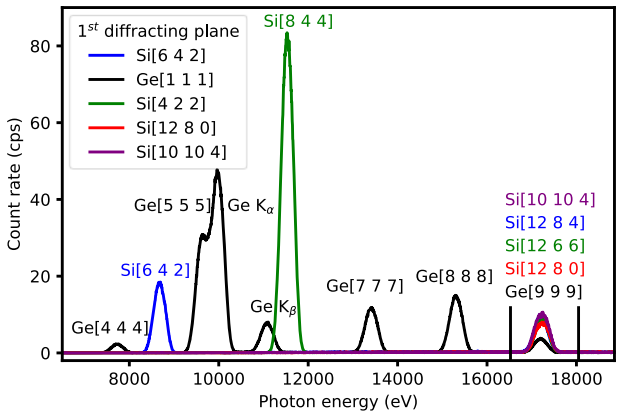



**FIG. 3.** Measured multi-channel-analyzer (MCA) spectra when the device is aligned at 17 115 eV using different crystal reflections. The ROI for U $L_3$-edge scan is highlighted by vertical black lines. As the crystal reflections contain harmonics, MCA spectra are denoted by the first diffracting plane in the legend (for interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article).

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harmonics up to 36 kV contrary to the Si[9 9 9]. It is important to note that the presence of harmonics can also be beneficial as they can serve as reference points, enabling the simultaneous measurement of $I_0$ and $I_t$.[51]

## III. RESULTS AND DISCUSSION

### A. Device performance

#### 1. Incident intensity

Figure 4 shows the intensity of the incident beam yielded by the selected reflections over the U and Pu $L_3$-edge energy range. The Ketek detector, with its larger active area, was used to optimize the collection efficiency in combination with a 16 × 2 $mm^2$ slit opening. For this comparison, the background and the diffraction peak within the region of interest of the MCA spectra were simultaneously fitted using linear and Gaussian functions, respectively. This approach was used to effectively remove any residual incoherent scattering from the measured signal during data processing.

The Ge[1 1 1] crystal is commonly used in symmetric configuration due to its versatility. It has been commonly used at the U $L_3$ edge[42,43,45] with the Ge[9 9 9] but yields only 224 cps in the present case. Using even reflections, such as Si[12 8 4] from Si[6 4 2], increases flux to 613 cps representing a 2.73 gain. However, Si[6 4 2] is mainly suited for the U $L_3$-edge and provides lower energy resolution for Np and Pu $L_3$-edges. Asymmetric configuration enables access to various *hkl* planes from standard crystal cuts, offering systematic flux optimization and improved energy resolution without relying on uncommon cuts. The Si[10 10 4] yields 708 cps at a low asymmetry angle of 2.5°. This is the most intense selected reflection, increasing the beam intensity by a factor of 3.16 compared to the Ge [9 9 9]. Increasing the asymmetry angle to 7.7°, 5.1°, or 10.1° yields 599, 544, or 500 cps for the Si[10 10 4], Si[12 6 6], and Si[12 8 0] reflections, respectively. Although the measured intensity is still higher than that of Ge[9 9 9], the Si[12 8 0] shows a count loss due to reduced collection efficiency from its lower Bragg and higher asymmetry angles, as confirmed by ray tracing simulations. Optimizing the detection system by using a larger detector would bring this reflection to the same level of the Si[10 10 4] with the asymmetry angle of 2.5°.

At the Pu $L_3$-edge, the gain of the asymmetric configuration is also significant. The Si[12 10 2] and Ge[10 10 8] reflections with asymmetry angles of 5.2° and 5.8°, respectively, yield the highest count rate. With 328 cps, this represents a gain of 1.74 on the 183 cps obtained with the Si[9 9 9], which was previously used for this energy range.[40] At higher asymmetry angles and lower Bragg angles, the intensity given by the Si[14 6 0] and Ge[14 6 4] reflections decreases but remains at around 269 and 255 cps, respectively. In comparison with the U $L_3$-edge, there is a noticeable decrease in incident intensity due to the reduced intensity of the x-ray tube profile as a fixed 24 W power was used.

Table I compares the intensity measured with the different reflections. Using the Ge[9 9 9] and Si[9 9 9] reflections as references from literature at the U and Pu $L_3$-edges, the table lists both measured and simulated intensity ratios (DE ratios). The model predicts the intensity ratio between Ge and Si reflections accurately within the uncertainty range. The integral reflectivity was found to be very sensitive to the Debye–Waller factor and the wafer thickness for Si reflections. Overall, the model successfully identifies the most intense reflection and provides correct order of magnitude estimates between odd reflections used in the literature and the stronger even reflections selected in this work.

The count rate obtained at the U $L_3$-edge, using the Si[10 10 4] reflection, is finally compared with the values reported in the literature in Table II. Direct comparison is difficult as the detected intensity depends on various instrumental settings, as can be seen from equation,[4] which differ from one device to another. In an attempt to give an overall indicator of the flux performance of these devices, the incident count rate is normalized by the tube power and yields the intensity efficiency. The HotXAS device achieves an intensity efficiency of about 29.5 cps $W^{-1}$, doubling what was obtained previously at the U $L_3$ edge. This increase is primarily due to the

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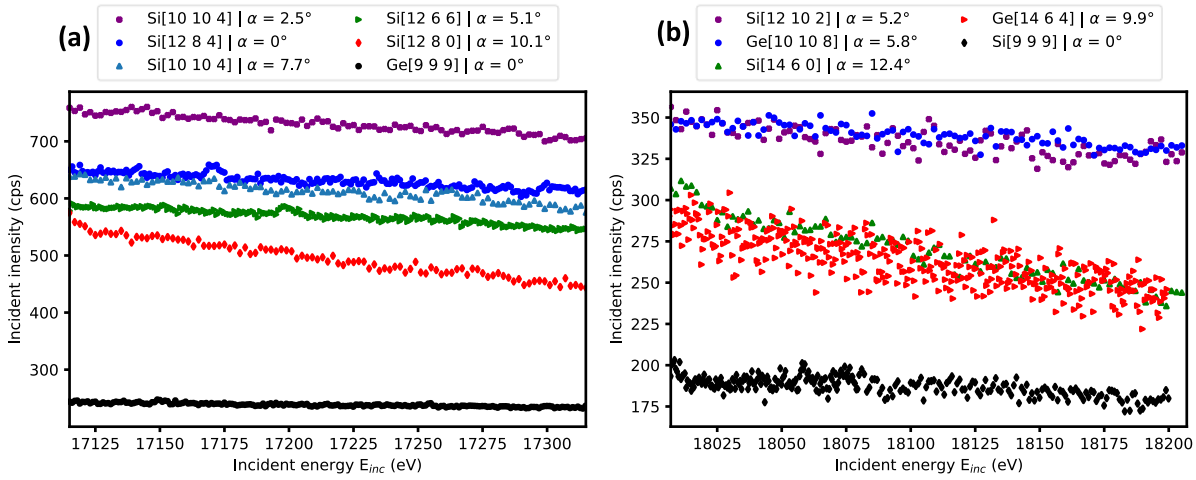



**FIG. 4.** Incident intensity measured the Ketek detector at the U $L_3$-edge (a) and Pu $L_3$-edge (b) for the selected reflections. The tube accelerating voltage and anode current were set at 24 kV and 1 mA, respectively. The same 300 $\mu$m thick aluminum filters were used.


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**TABLE II.** Comparison of count rate reported at the U $L_3$-edge normalized by the tube power. The reported instrumental settings are listed, including the crystal reflection, tube accelerating voltage, anode material and current, crystal solid angle, and detector thickness. The collection efficiency is about 100% for all experiments. The count rates obtained with the HotXAS device were subtracted from background counts. This assumption is also made in all reported values in the literature, or the contribution is assumed to be very low, given the detectors' low opening area and good energy resolution. The filter thicknesses are also assumed to be similar. No reported value of the count rate was found for the other works at the U[43,44] and Pu[45] $L_3$-edges.

| Edge | $E$ (eV) | Refs. | $hkl$ | $V$ (kV) | $I$ (mA) | Target | $SA_c$ (msr) | Detection area ($mm^2$) | Detector | $I_0$ (cps) | $I_0/P$ (cps $W^{-1}$) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| U $L_3$ | 17 166 | 42 | Ge[9 9 9] | 30 | 10 | Ag | 11 | >7 | 1 mm CdTe | 700 | 2.3 |
| | | 45 | Ge[9 9 9] | 30 | 10 | Ag | 11 | >70 | 500 $\mu$m Si | 4000 | 13.3 |
| | | This work | Si[10 10 4] | 24 | 1 | Pd | 32 | 28 | 450 $\mu$m Si | 708 | 29.5 |

increased reflectivity achieved through the use of the Si[10 10 4] reflection, as well as the x-ray tube's larger opening angle seen in the crystal solid angle $SA_c$.

### 2. Resolution

In order to provide a comprehensive description of the HotXAS device's performance, the instrumental energy resolution was evaluated for each of the reflections measured in this study. The geometrical contribution, $\Delta E_g$, was estimated using x-ray tracing simulations with the ray tracer application,[30] while the intrinsic crystal resolution of the bent crystal, $\Delta E_c$, was computed using the TBCA package of the PyTTE software[48] to simulate the diffraction curve of the strip bent crystals. The results of the simulations are given in Table III alongside the total convoluted instrumental energy resolution, $\Delta E_i$.

The geometrical resolution is ~2 eV or less for all the selected reflections. As expected, the asymmetric configuration considerably reduces the Johann error for all the selected reflections compared to their symmetric counterparts. Note the geometrical energy resolution of 0.2 eV obtained with the Si[12 10 2] reflection, which yields an incidence angle of 89.8° ($\Theta M$) with an asymmetry angle of 5.2°. The main contributor to the total instrumental resolution is the intrinsic resolution for Si reflections, which is about 2 eV. For the Ge crystals, the geometrical resolution is higher, with the intrinsic resolution being only around 0.4 eV, given the lower penetrating depth of photons in the strained wafer compared to Si. Combining excellent geometrical and intrinsic resolutions, the Ge[9 9 9] and Ge[10 10 8] reflections yield exceptional instrumental resolution of 0.6 and 0.8 eV at 17 166 and 18 057 eV, respectively. Nonetheless, all the selected reflections also yield an instrumental resolution of ~2.5 eV, which is suitable for measurements at the U and Pu $L_3$-edges, where the core hole broadening is about 8.2 and 8.7 eV, respectively. At this energy, the resolving power ($E/\Delta E$) is ~7000, which is very close to the standards of synchrotron beamlines at 10 000. In Fig. S4 of the supplementary material, the reader will find XANES spectra collected at the Zn and Cu K-edges in reference foils in symmetric

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**TABLE III.** Instrumental resolution of the HotXAS device at U and Pu $L_3$-edges for all the reflections measured in this work. Commonly used reflections are highlighted in boldface. $\Delta E_g$ and $\Delta E_c$ are given as the full width at half maximum of the geometrical response function and diffraction curves of the crystals.

| Reflection | $\Theta_B$ (°) | $\alpha$ (°) | $\Theta_M$ (°) | $\Delta E_g$ sym (eV) | $\Delta E_g$ asym (eV) | $\Delta E_c$ (eV) | $\Delta E_i$ (eV) | $\Delta E/E$ |
|---|---|---|---|---|---|---|---|---|
| **U $L_3$: 17 166 eV** | | | | | | | | |
| Si[12 8 0] | 73.6 | 10.1 | 83.7 | 4.4 | 1.7 | 1.5 | 2.3 | $1.3 \times 10^{-4}$ |
| Si[12 6 6] | 77.6 | 5.1 | 82.8 | 2.5 | 1.5 | 2.1 | 2.6 | $1.5 \times 10^{-4}$ |
| Si[10 10 4] | 77.8 | 2.5 | 80.3 | 2.4 | 1.9 | 2.1 | 2.8 | $1.6 \times 10^{-4}$ |
| Si[10 10 4] | 77.8 | 7.7 | 85.5 | 2.4 | 0.9 | 2.1 | 2.3 | $1.4 \times 10^{-4}$ |
| Si[12 8 4] | 84.5 | 0 | 84.5 | 0.5 | 0.5 | 2.2 | 2.3 | $1.3 \times 10^{-4}$ |
| ***Ge[9 9 9]*** | ***84.3*** | ***0*** | ***84.3*** | ***0.5*** | ***0.5*** | ***0.4*** | ***0.6*** | $\boldsymbol{3.6 \times 10^{-5}}$ |
| **Pu $L_3$: 18 057 eV** | | | | | | | | |
| Ge[14 6 4] | 72.9 | 9.9 | 82.8 | 5.1 | 2.1 | 0.3 | 2.1 | $1.2 \times 10^{-4}$ |
| Ge[10 10 8] | 80.3 | 5.8 | 86.1 | 1.6 | 0.7 | 0.4 | 0.8 | $4.5 \times 10^{-5}$ |
| Si[14 6 0] | 74.4 | 12.4 | 86.8 | 4.2 | 0.9 | 1.7 | 2.0 | $1.1 \times 10^{-4}$ |
| Si[12 10 2] | 84.7 | 5.2 | 89.8 | 0.5 | 0.2 | 2.7 | 2.7 | $1.5 \times 10^{-4}$ |
| ***Si[9 9 9]*** | ***80.2*** | ***0*** | ***80.2*** | ***1.6*** | ***1.6*** | ***2.4*** | ***2.8*** | $\boldsymbol{1.6 \times 10^{-4}}$ |


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configuration, where the laboratory results are in good agreement with the synchrotron data.

## B. Applications

To showcase the capabilities of the HotXAS device for actinide speciation in nuclear fuels, two sets of XANES spectra were collected. The feasibility of measuring EXAFS with the current x-ray source setting was also evaluated.

### 1. Speciation of Zr

First, the device was tested on inactive materials by measuring the Zr K-edge (17 998 eV) in reference compounds, in order to validate its performance in the close energy range of the Pu K-edge (18 057 eV). Zr is used here as a surrogate element of Pu. Ge[10 10 8] reflection was used in combination with the Si detector (acquisition parameters for all the compounds measured in this work are given in Tables S5 and S6 of the supplementary material).

As shown in Fig. 5, the XANES spectrum of the metallic foil exhibits very good agreement with the synchrotron data. With an incident angle of 88° at 17 998 eV, the Ge[10 10 8] reflection provides a 0.8 eV instrumental resolution being below the 4.8 eV core hole broadening. All the spectral features are resolved. Given in Fig. S5 of the supplementary material, the absorbance difference between the synchrotron and laboratory spectra shows no differences except for the shoulder in the edge jump of the Zr foil spectra where the HotXAS device better resolves this feature due to the 0.5 eV energy step used as opposed to 1 eV for the synchrotron data. Absorbance differences also reveal a slightly lower intensity in the white line of the $ZrO_2$ spectrum of the HotXAS device attributed to a difference of thickness between the $ZrO_2$ samples. The effect of background, accounting for possible additional thickness effects,[49] was also evaluated in Fig. S5 for both compounds, but the spectral differences show no differences between the data processing methods: fitting the ROI of the MCA spectra to remove the residual background, as opposed to summing the ROI directly. At 24 W, the signal-to-noise ratio is reasonable with 27 h of measurements using the Ge[10 10 8] reflection. The shift in edge position and change in coordination geometry are clearly visible when moving from the metallic foil (+0) to the $ZrO_2$ compound (+IV). These results are obviously promising for the speciation of Pu in MOX nuclear fuel samples.

The feasibility of analyzing diluted samples with the current beam intensity was assessed in Fig. S6 of the supplementary material by simulating the Pu $L_3$ XANES spectra, accounting for the Poisson statistics associated with the Beer–Lambert law. Considering samples compatible with transmission mode, acquisition time increases exponentially as Pu content decreases. Concentrated samples (Pu > 40%) can be measured under 40 h; however, at a Pu content of 10%, ~700 h are required to achieve reasonable statistics. With the present x-ray source power settings, the practical lower limit appears to fall between 10% and 20% Pu, where acquisition time is roughly 1 week, an acceptable duration for laboratory operations that enables XANES characterization of SFR fuels. This preliminary assessment will soon be verified through direct characterization of MOX samples with varying Pu content.

### 2. Characterization of U

The second set of measurements was performed on $U_{1-y}Ce_yO_{2-x}$ samples, which were manufactured and characterized in a previous study by Massonnet *et al.*[54] and were used here as a surrogate material for $(U,Pu)O_2$. In this study, we measured the uranium oxidation state in relation to the manufacturing conditions for four samples containing 14% Ce. For these measurements, the Si[12 8 4] reflection was used to maximize the collection efficiency, given the 5 mm diameter of the samples. In combination with the CdTe detector, the incident count rate was ~1300 cps.

Figure 6 plots the U $L_3$-edge XANES spectra for the four samples measured with the HotXAS device, alongside synchrotron

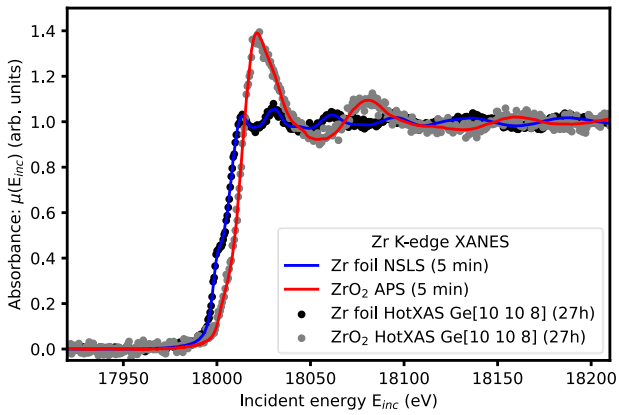



**FIG. 5.** Zr K-edge spectra measured in the metallic reference foil and in the $ZrO_2$ compound with the HotXAS device (individual scans merged and calibrated in Bragg space) and compared to the synchrotron data collected at the National Synchrotron Light Source (NSLS) and the Advanced Photon Source (APS), available online from the Hephaestus Software[52] and the XAS database.[53] The acquisition time for the synchrotron data is given as an estimate. The acquisition parameters for the HotXAS data are given in Tables S5 an S6 of the supplementary material.

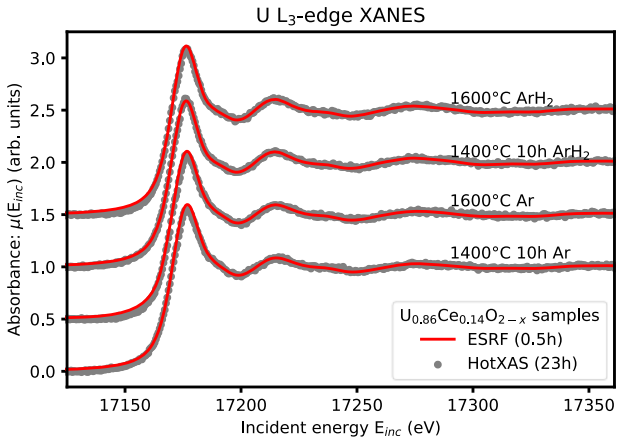



**FIG. 6.** U $L_3$ XANES spectra of $U_{1-y}Ce_yO_{2-x}$ nuclear fuel samples and compared to the synchrotron data. The acquisition time of the synchrotron data is given as an estimate. The acquisition parameters are given in Tables S5 and S6 of the supplementary material for the HotXAS data (individual scans merged and calibrated in Bragg space).

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measurements. Thanks to the instrument's 2.6 eV energy resolution, all spectral features seen in the synchrotron data are clearly resolved. However, there is a slight mismatch in the pre-edge and white line regions, which is evident from the absorbance differences presented in Fig. S7 of the supplementary material. This arises from background due to the limited pre-edge range offered by the Si[12 8 4] reflection, which yields an 86.5° Bragg angle at 17 115 eV (appearing as 17 125 eV after calibration); at this angle, the tube and sample holder nearly collide. Improvements are being developed to solve this issue, i.e., using the Si[10 10 4] reflection in asymmetric configuration, which yields a 78.6° Bragg angle at the same energy.

Although the noise in the spectrum obtained from 23-h measurements is higher than in the synchrotron measurement (0.5 eV energy steps were used on the entire energy range, and SNR can already be increased with the current device settings by optimizing the acquisition parameters), the data are, nevertheless, suitable for fingerprinting and extraction of the oxidation state. The evolution of the statistics as a function of acquisition time was evaluated and is reported in Fig. S8 of the supplementary material.

Three hours of measurements were found to yield reasonable statistics. Increasing the acquisition time beyond 24 h did not yield a significant improvement. Following standard XAS protocols, the total measurement was divided into several shorter scans. Device stability was verified by analyzing changes between individual scans of the $UO_2$ reference (cf. same Fig. S8 of the supplementary material). With proper ventilation of the shielding unit to remove heat from the x-ray source, no shift or intensity variation in features was observed, confirming adequate heat dissipation and thermal stability of the crystal. Significant heat generation within the crystal due to x-ray irradiation is also unlikely given the low power.

To extract uranium oxidation states, $UO_2$(IV) and $U_4O_9$(IV/V) served as reference compounds for linear combination analysis (LCA) using the Athena software.[52] The reference spectra measured on the device are given in Fig. S9 of the supplementary material. The results of the LCA are presented in Table IV and are compared with the synchrotron results. The measured U OS by Massonnet *et al.* in HERFD mode ranges from 4.04 to 4.26. In standard transmission mode, synchrotron data suggest slight oxidation of the samples, with U OS ranging from 4.07 to 4.30. These differences can be attributed to oxidation during storage between the two synchrotron experiments (a 2-year difference). The data collected with the HotXAS device follows the trend of the synchrotron experiments, with extracted U OS ranging from 4.00 to 4.25. For each sample, the laboratory measure is close to the synchrotron results within the uncertainty range. Considering the 24 W tube power, this uncertainty is about 0.03 in 24 h.

The feasibility of measuring EXAFS data for concentrated actinide samples with the current x-ray source settings was also assessed in Fig. S10 of the supplementary material through spectrum simulations. Acquisition times are significantly longer than those required for XANES, primarily due to the increasing integration time needed to compensate for the dumping of EXAFS oscillations at higher wavenumbers. Nevertheless, EXAFS data with acceptable statistics can be collected within one week up to 13.5 $Å^{-1}$ at the U $L_3$-edge, and within 2 weeks at the Pu $L_3$-edge. This provides an excellent foundation for routine XAS characterizations of MOX sample within nuclear research facilities.

## IV. CONCLUSION

The development of the HotXAS device, a new laboratory XAS setup dedicated to the characterization of MOX nuclear fuels, has been reported. The device was designed to deal with radioactive samples studied in nuclear fuel research facilities. It offers easy and unlimited access to an x-ray beam, as opposed to a synchrotron beamline.

The device's performance was optimized for scanning the energy ranges of U and Pu $L_3$-edges by exploring the Johann asymmetric configuration. Operating in this configuration was shown to provide a systematic way of optimizing beam intensity and energy resolution simultaneously, by easily accessing high-reflectivity reflections with a limited number of crystals. Beam intensity increased by a factor of about 3 with Si[10 10 4] and by a factor of about 2 with Ge[10 10 8], compared to Ge[9 9 9] and Si[9 9 9] used previously in symmetric configuration at the U and Pu $L_3$-edges, respectively, while maintaining excellent energy resolution. It also provides a means of eliminating lower-order harmonics, which are a systematic challenge for the detection system in high-energy XAS experiments. The application of asymmetric configuration for intensity optimization holds promises for other actinides, 4 d, and rare earth metals.

With a low tube power of 24 W and a moderate acquisition time of 23 h, the device was used to determine the uranium oxidation state in $U_{0.86}Ce_{0.14}O_2$ samples with a 0.03 error. Used as a spectroscopic

**TABLE IV.** Measured uranium oxidation states in $U_{1-y}Ce_yO_{2-x}$ samples with the HotXAS device and compared to synchrotron results at U $L_3$-edge (analysis of the EXAFS data from the study of Massonet *et al.*) alongside reported U OS at U $M_4$ edge in HERFD mode. The errors are reported in parentheses, and the fit is performed from −10 to +30 eV relative to the edge for the U $L_3$ data.

| Sintering conditions | Atm | $UO_2$ | $U_4O_9$ | U $L_3$ HotXAS | U $L_3$ (ROBL)[54] | HERFD $M_4$ (Soleil)[54] |
|---|---|---|---|---|---|---|
| 1600 °C; 0 h | $Ar/H_2$ | 95 (3) | 5 (3) | 4.03 (0.01) | 4.08 (0.01) | 4.04 (0.01) |
| 1400 °C; 10 h | $Ar/H_2$ | 100 (0) | 0 (0) | 4.00 (0) | 4.07 (0.02) | 4.06 (0.01) |
| 1600 °C; 0 h | Ar | 58 (5) | 42 (5) | 4.21 (0.03) | 4.21 (0.01) | 4.20 (0.01) |
| 1400 °C; 10 h | Ar | 49 (4) | 51 (4) | 4.25 (0.02) | 4.30 (0.01) | 4.26 (0.01) |

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analog of Pu, speciation of Zr was also demonstrated by measuring a Zr metal foil and a $ZrO_2$ compound. The expected Pu dilution limit is about 10–20%. Spectrum simulations also show that the device can provide structural information from EXAFS spectra within a week for concentrated samples. The HotXAS device is an alternative to synchrotrons for the routine characterization of actinide samples, with potential applications in many fields of nuclear fuel and actinide research, including manufacturing of advanced nuclear fuels and reprocessing and the long-term disposal of nuclear waste.

## SUPPLEMENTARY MATERIAL

The supplementary material contains the detailed information on the following specific aspects: Figures S1 and S2 present collection efficiency vs Bragg angle for symmetric and asymmetric setups. Figure S3 covers background as a function of slit opening/crystal reflection and its impact on SNR. Radioactive background counts for different Pu contents are in Tables S2–S4 with the corresponding model parameter presented in Table S1. Figure S4 displays Zn K and Cu K-edge XANES spectra from reference foils. Spectral difference between synchrotron and laboratory measurements at Zr K- and U $L_3$-edges are in Figs. S5 and S7. Device settings and acquisition parameters are listed in Tables S5 and S6. Spectra simulation statistics appear in Fig. S6 (Pu $L_3$ XANES) and Fig. S10 (EXAFS of concentrated samples). Figure S8 experimentally examines the U $L_3$ XANES spectra statistics of the $UO_2$ reference with varying acquisition time and examines the device stability under extended measurements. Figure S9 shows U $L_3$ reference spectra used for uranium oxidation state determination in $(U,Ce)O_2$ samples. Spectra simulation statistics are shown in Fig. S9 (Pu $L_3$ XANES) and Fig. S10 (EXAFS of concentrated samples).

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

**S. Orlat**: Conceptualization (equal); Formal analysis (equal); Writing – original draft (equal). **P. Martin**: Conceptualization (lead); Resources (lead); Supervision (equal); Writing – review & editing (equal). **P. Moisy**: Funding acquisition (supporting); Supervision (equal); Writing – review & editing (equal). **F. Tuomisto**: Funding acquisition (supporting); Resources (supporting); Supervision (supporting). **R. Bes**: Conceptualization (equal); Software (equal); Supervision (equal); Writing – review & editing (equal).

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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